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Inventor(s)	LI; Jingchen et al.

METHOD AND APPARATUS FOR CONTROLLING MOBILE ROBOT, MOBILE ROBOT, AND STORAGE MEDIUM

Abstract

A method for controlling a mobile robot includes controlling the mobile robot to be in an initial standing state; and controlling the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among a first swinging leg set of the mobile robot and a second swinging leg set of the mobile robot are horizontally closed together, wherein at least one of the first swinging leg set and the second swinging leg set includes swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane.

Inventors: **LI; Jingchen (Shenzhen, CN), ZHANG; Yufeng (Shenzhen, CN), ZHANG; Dongsheng (Shenzhen, CN), LI; Xiong (Shenzhen, CN), WANG; Shuai (Shenzhen, CN), ZHOU; Qinqin (Shenzhen, CN), SHENG; Minhua (Shenzhen, CN)**

Applicant: **Tencent Technology (Shenzhen) Company Limited** (Shenzhen, CN)

Family ID: **93141391**

Assignee: **Name Tencent Technology (Shenzhen) Company Limited (Shenzhen, CN)**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is a continuation application of International Application No. PCT/CN2023/131455 filed on Nov. 14, 2023, which claims priority to Chinese Patent Application No. 202310475906.8 filed with the China National Intellectual Property Administration on Apr. 25, 2023, the disclosures of each being incorporated by reference herein in their entireties.
FIELD
[0002] The disclosure relates to the field of robots, and to a method and an apparatus for controlling a mobile robot, a mobile robot, and a storage medium.
BACKGROUND
[0003] To ensure higher operating performance of a robot, the robot may be provided with a plurality of mechanical assemblies, such as robot legs, robot arms, and a torso structure. A robot provided with many mechanical assemblies may have a large volume. Reducing the volume of a robot in a non-operating state is accordingly a technical problem that to be resolved.
SUMMARY
[0004] According to an aspect of the disclosure, a method for controlling a mobile robot that includes a first swinging leg set and a second swinging leg set, wherein at least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs, the first swinging leg set and the second swinging leg set are arranged side by side, and a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane, includes controlling the mobile robot to be in an initial standing state; and controlling the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among the first swinging leg set and the second swinging leg set are horizontally closed together.
[0005] According to an aspect of the disclosure, an apparatus for controlling a mobile robot, includes at least one memory configured to store computer program code; and at least one processor configured to read the program code and operate as instructed by the program code, wherein the mobile robot includes a first swinging leg set and a second swinging leg set, wherein at least one of the first swinging leg set and the second swinging leg set include a plurality of swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane, and wherein the program code includes first control code configured to cause at least one of the at least on processor to control the mobile robot to be in an initial standing state; and second control code configured to cause at least one of the at least one processor to control the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among the first swinging leg set and the second swinging leg set are horizontally closed together.
[0006] According to an aspect of the disclosure, a non-transitory computer-readable storage medium, storing computer code which, when executed by at least one processor, causes the at least

one processor to at least control a mobile robot to be in an initial standing state; and control the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among a first swinging leg set of the mobile robot and a second swinging leg set of the mobile robot are horizontally closed together, wherein at least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] To describe the technical solutions of some embodiments of this disclosure more clearly, the following briefly introduces the accompanying drawings for describing some embodiments. The accompanying drawings in the following description show only some embodiments of the disclosure, and a person of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts. In addition, one of ordinary skill would understand that aspects of some embodiments may be combined together or implemented alone.

[0008] FIG. **1** is a side view of a mobile robot according to some embodiments.

[0009] FIG. **2** is a side view of a mobile robot according to some embodiments.

[0010] FIG. **3** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0011] FIG. **4** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0012] FIG. **5** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0013] FIG. **6** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0014] FIG. **7** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0015] FIG. **8** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0016] FIG. **9** is a schematic diagram of a sideway swinging state according to some embodiments.

[0017] FIG. **10** is a schematic diagram of a sideway swinging state and a bent state according to some embodiments.

[0018] FIG. **11** is a schematic diagram of a torso structure in a sideway swinging state and a bent state, with a motion arm being in a first extended state according to some embodiments.

[0019] FIG. **12** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0020] FIG. **13** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0021] FIG. **14** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0022] FIG. **15** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0023] FIG. **16** is a flowchart of a method for controlling a mobile robot according to some embodiments.

[0024] FIG. **17** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.

[0025] FIG. **18** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0026] FIG. **19** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.
[0027] FIG. **20** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0028] FIG. **21** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.
[0029] FIG. **22** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0030] FIG. **23** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.
[0031] FIG. **24** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0032] FIG. **25** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0033] FIG. **26** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.
[0034] FIG. **27** is a flowchart of a method for controlling a mobile robot according to some embodiments.
[0035] FIG. **28** is a schematic diagram of a method for controlling a mobile robot according to some embodiments.
[0036] FIG. **29** is a schematic diagram of an apparatus for controlling a mobile robot according to some embodiments.
[0037] FIG. **30** is a schematic diagram of an apparatus for controlling a mobile robot according to some embodiments.
[0038] FIG. **31** is a block diagram of a mobile robot according to some embodiments.

DESCRIPTION OF EMBODIMENTS

[0039] To make the objectives, technical solutions, and advantages of the present disclosure clearer, the following further describes the present disclosure in detail with reference to the accompanying drawings. The described embodiments are not to be construed as a limitation to the present disclosure. All other embodiments obtained by a person of ordinary skill in the art without creative efforts shall fall within the protection scope of the present disclosure.
[0040] In the following descriptions, related “some embodiments” describe a subset of all possible embodiments. However, it may be understood that the “some embodiments” may be the same subset or different subsets of all the possible embodiments, and may be combined with each other without conflict. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. For example, the phrase “at least one of A, B, and C” includes within its scope “only A”, “only B”, “only C”, “A and B”, “B and C”, “A and C” and “all of A, B, and C.”
[0041] A structure of a mobile robot involved in some embodiments is described.
[0042] With reference to FIG. **1** and FIG. **2**, it may be observed that the mobile robot provided in some embodiments includes at least three swinging legs. The at least three swinging legs are arranged side by side and rotation axes are located on the same vertical plane. Arranged side by side is a state in which projections of the at least three swinging legs of the mobile robot in a first direction do not overlap. That the rotation axes are located on the same vertical plane refers to a state in which projections of the at least three swinging legs of the mobile robot in a second direction do not overlap. The first direction refers to an orientation of a front side or a back side of the mobile robot. The second direction refers to an orientation of a lateral side of the mobile robot.

In some embodiments, rotation axes of at least two of the at least three swinging legs are coaxial. In some embodiments, rotation axes of the at least three swinging legs are not coaxial.
[0043] In some embodiments, the at least three swinging legs are divided to a first swinging leg set and a second swinging leg set. In some embodiments, the first swinging leg set includes a plurality of first swinging legs, the second swinging leg set includes one second swinging leg, at least two first swinging legs of the plurality of first swinging legs are respectively located on two sides of a central axis of the mobile robot, and the second swinging leg is located on the central axis of the mobile robot. In some embodiments, the first swinging leg set includes one first swinging leg, the second swinging leg set includes a plurality of second swinging legs, at least two second swinging legs of the plurality of second swinging legs are respectively located on two sides of a central axis of the mobile robot, and the first swinging leg is located on the central axis of the mobile robot.
[0044] In some embodiments, the at least three swinging legs are divided to a first swinging leg set and a second swinging leg set. In some embodiments, the first swinging leg set includes a plurality of first swinging legs, the second swinging leg set includes a plurality of second swinging legs, and the plurality means greater than or equal to two.
[0045] With reference to FIG. **1** and FIG. **2**, FIG. **1** and FIG. **2** are side views of a mobile robot according to some embodiments. The mobile robot includes a first swinging leg set **10** and a second swinging leg set **20**. The first swinging leg set **10** includes a plurality of first swinging legs **11**, and the second swinging leg set **20** includes a plurality of second swinging legs **21**. At least two of the plurality of first swinging legs **11** are respectively located on two sides of a central axis of the mobile robot, and at least two of the plurality of second swinging legs **21** are respectively located on the two sides of the central axis of the mobile robot. The plurality of first swinging legs **11** and the plurality of second swinging legs **21** are distributed side by side. In a case that the foregoing distribution condition is satisfied, in some embodiments, the plurality of first swinging legs **11** and the plurality of second swinging legs **21** are alternately distributed one by one. In some embodiments, the plurality of first swinging legs **11** are distributed on two sides of the plurality of second swinging legs **21**.
[0046] An example in which the first swinging leg set **10** is a leg set A, and the second swinging leg set **20** is a leg set B. The plurality of first swinging legs **11** and the plurality of second swinging legs **21** may be distributed side by side as follows: A1, B1, B2, and A2 (a distribution case 1), A1, B1, A2, and B2 (a distribution case 2), A1, A2, B1, A3, B2, and B3 (a distribution case 3), A1, A2, B1, B2, B3, and A3 (a distribution case 4), or the like. Likewise, more or fewer swinging legs may be arranged side by side in a similar manner.
[0047] With reference to FIG. **1** and FIG. **2**, FIG. **1** and FIG. **2** show that the first swinging leg set **10** includes two first swinging legs (outer legs) **11**, and the second swinging leg set **20** includes two second swinging legs (inner legs) **21**. The two first swinging legs **11** are symmetrically distributed along the central axis of the mobile robot, and the two second swinging legs **21** are symmetrically distributed along the central axis of the mobile robot. A distance between each of the first swinging legs **11** and the central axis is greater than a distance between each of the second swinging legs **21** and the central axis.
[0048] During travelling of the mobile robot, the first swinging leg set **10** and the second swinging leg set **20** may travel with crossed steps. The first swinging leg set **10** and the second swinging leg set **20** may travel alternately in a front-rear direction. In an exemplary travelling process, the second swinging leg set **20** is swung to a first stop point by using the first swinging leg set **10** as support legs, and the first swinging leg set **10** is swung to a second stop point by using the second swinging leg set **20** as support legs.
[0049] In an initial posture, the plurality of first swinging legs **11** may serve as front legs and are in contact with the ground, and the plurality of second swinging legs **21** serve as rear legs and are in contact with the ground. A projection of a center of gravity of the robot may be located between geometric figures formed by contact points of the front and rear legs and the ground. The plurality

of second swinging legs **21** are swung to the first stop point by using the plurality of first swinging legs **11** as support legs, and at the same time, the center of gravity of the robot is controlled to move forward. After the plurality of second swinging legs **21** are swung to the first stop point, the center of gravity of the robot is controlled to be located between the geometric figures formed by the contact points of the front and rear legs and the ground again. The plurality of first swinging legs **11** are swung to the second stop point by using the plurality of second swinging legs **21** as support legs, and at the same time, the center of gravity of the robot is controlled to move forward. After the plurality of second swinging legs **21** are swung to the second stop point, the center of gravity of the robot is controlled to be located between the geometric figures formed by the contact points of the front and rear legs and the ground again.

[0050] During the swinging of the swinging legs of the robot, the swinging legs are controlled to stretch or retract. In a case that centers of gravity of the swinging legs are located behind the center of gravity of the mobile robot, the swinging legs may be controlled to contract. In a case that the centers of gravity of the swinging legs are located before the center of gravity of the mobile robot, the swinging legs are controlled to stretch.

[0051] According to the foregoing mobile robot, the first swinging leg set and the second swinging leg set are arranged, the first swinging leg set includes a plurality of first swinging legs, the second swinging leg set includes a plurality of second swinging legs, at least two of the plurality of first swinging legs are respectively located on the two sides of the central axis of the mobile robot, and at least two of the plurality of second swinging legs are respectively located on the two sides of the central axis of the mobile robot, and the plurality of first swinging legs and the plurality of second swinging legs are distributed side by side. Static stability of the mobile robot in a standing posture can be achieved, and a position of the center of gravity of the mobile robot may not be dynamically adjusted. The foregoing mobile robot can travel with crossed steps. During the travelling, the mobile robot does not have a problem regarding balance in a rolling direction. The rolling direction is a direction perpendicular to a travelling direction.

[0052] In some embodiments, the plurality of first swinging legs **11** are rotatably connected to a first swinging rotation axis, and the first swinging rotation axis is perpendicular to a travelling direction of the mobile robot. In some embodiments, the first swinging rotation axis is located at a position such as a hip, a waist, or a head top of the mobile robot. In some embodiments, the plurality of second swinging legs **21** are rotatably connected to a second swinging rotation axis, and the second swinging rotation axis is perpendicular to the travelling direction of the mobile robot. In some embodiments, the second swinging rotation axis is located at a position such as a hip, a waist, or a head top of the mobile robot.

[0053] In some embodiments, the first swinging rotation axis is located at a hip of the mobile robot. With reference to FIG. **1**, the first swinging rotation axis is a first hip rotation axis **1**. When the mobile robot stands on a horizontal reference plane, the first hip rotation axis **1** extends horizontally. The plurality of first swinging legs **11** are rotatably connected to the first hip rotation axis **1**, and any two of the plurality of first swinging legs **11** are parallel.

[0054] In some embodiments, the second swinging rotation axis is located at a hip of the mobile robot. With reference to FIG. **1**, the second swinging rotation axis is a second hip rotation axis **2**, and when the mobile robot stands on a horizontal reference plane, the second hip rotation axis **2** extends horizontally. The plurality of second swinging legs **21** are rotatably connected to the second hip rotation axis **2**, and any two of the plurality of second swinging legs **21** are parallel.

[0055] In some embodiments, the first hip rotation axis **1** and the second hip rotation axis **2** are coaxial. FIG. **1** and FIG. **2** show a case in which the first hip rotation axis **1** and the second hip rotation axis **2** are coaxial and in the same vertical plane.

[0056] In some embodiments, the mobile robot further includes a first rotating motor and a second rotating motor. The first rotating motor is configured to drive the plurality of first swinging legs **11** in the first swinging leg set **10** to interactively rotate about the first hip rotation axis **1**. The second rotating motor is configured to drive the plurality of second swinging legs **21** in the second swinging leg set **20** to interactively rotate about the second hip rotation axis **2**.

[0057] In some embodiments, the mobile robot further includes a third rotating motor corresponding to the first swinging leg **11** and a fourth rotating motor corresponding to the second swinging leg **21**. The third rotating motor is configured to drive the first swinging leg **11** to rotate about the first hip rotation axis **1**. The fourth rotating motor is configured to drive the second swinging leg **21** to rotate about the second hip rotation axis **2**. In some embodiments, the plurality of third rotating motors respectively corresponding to the plurality of first swinging legs **11** can control the plurality of first swinging legs **11** to rotate associatively or independently. In some embodiments, the plurality of fourth rotating motors respectively corresponding to the plurality of second swinging legs **21** can control the plurality of second swinging legs **21** to rotate associatively or independently.

[0058] With reference to FIG. **1** and FIG. **2**, the first swinging leg **11** includes a first robotic thigh **111** and a first robotic calf **112**. The first robotic thigh **111** is connected to the first robotic calf **112** through sleeving. In some embodiments, in the sleeved state, the first robotic thigh **111** is nested inside the first robotic calf **112**. During stretching or retraction, the first robotic thigh **111** is stretched or retracted in a sleeving direction. In some embodiments, in the sleeved state, the first robotic calf **112** is nested inside first robotic thigh **111**. During stretching or retraction, the first robotic calf **112** is stretched or retracted in the sleeving direction (a case shown in FIG. **1** and FIG. **2**). In some embodiments, in the sleeved state, the first robotic thigh **111** and the first robotic calf **112** are nested inside the intermediate piece. During stretching or retraction, the first robotic thigh **111** and the first robotic calf **112** are stretched or retracted in the sleeving direction.

[0059] The second swinging leg **21** includes a second robotic thigh **211** and a second robotic calf **212**. The second robotic thigh **211** is connected to the second robotic calf **212** through sleeving. In some embodiments, in the sleeved state, the second robotic thigh **211** is nested inside the second robotic calf **212**. During stretching or retraction, the second robotic thigh **211** is stretched or retracted in a sleeving direction. In some embodiments, in the sleeved state, the second robotic calf **212** is nested inside the second robotic thigh **211**. During stretching or retraction, the second robotic calf **212** is stretched or retracted in the sleeving direction (a case shown in FIG. **1** and FIG. **2**). In some embodiments, in the sleeved state, the second robotic thigh **211** and the second robotic calf **212** are nested inside the intermediate piece. During stretching or retraction, the second robotic thigh **211** and the second robotic calf **212** are stretched or retracted in the sleeving direction.

[0060] In some embodiments, the mobile robot further includes a first stretching and retraction motor corresponding to the first swinging leg **11** and a second stretching and retraction motor corresponding to the second swinging leg **21**. The first stretching and retraction motor is configured to drive the first swinging leg **11** to stretch or retract in the sleeving direction. The second stretching and retraction motor is configured to drive the second swinging leg **21** to stretch or retract in the sleeving direction. In some embodiments, the first stretching and retraction motor is a first linear motor. In some embodiments, the first stretching and retraction motor is a motor designed to implement linear transmission through a screw nut. In some embodiments, the second stretching and retraction motor is a second linear motor. In some embodiments, the second stretching and retraction motor is a motor designed to implement linear transmission through the screw nut.

[0061] In some embodiments, a plurality of first stretching and retraction motors respectively corresponding to the plurality of first swinging legs **11** can control the plurality of first swinging legs **11** to stretch or retract associatively or independently. In some embodiments, a plurality of second stretching and retraction motors respectively corresponding to the plurality of second swinging legs **21** can control the plurality of second swinging legs **21** to stretch or retract associatively or independently.

[0062] In some embodiments, the first swinging leg **11** includes the first robotic thigh **111** and the first robotic calf **112**. The first robotic thigh **111** and the first robotic calf **112** are in a rotation connection through a first knee joint rotation axis. The first knee joint rotation axis supports increasing or decreasing an included angle between the first robotic thigh **111** and the first robotic calf **112**. The second swinging leg **21** includes the second robotic thigh **211** and the second robotic calf **212**. The second robotic thigh **211** and the second robotic calf **212** are in a rotation connection through a second knee joint rotation axis. The second knee joint rotation axis supports increasing or decreasing an included angle between the second robotic thigh **211** and the second robotic calf **212**.

[0063] In some embodiments, the first swinging leg set **10** includes the plurality of first swinging legs **11**. Each of the first swinging legs **11** includes a first leg assembly and a first wheel **113** located on an end of the first leg assembly. The second swinging leg set **20** includes the plurality of second swinging legs **21**. The second swinging leg **21** includes a second leg assembly and a second wheel **213** located on an end of the second leg assembly. In some embodiments, the first wheel **113** is a wheel having a multi-directional degree of freedom. The first wheel **113** supports rotation in any direction. In some embodiments, the second wheel **213** is a wheel having a multi-directional degree of freedom, and the second wheel **213** supports rotation in any direction. With reference to FIG. **1** and FIG. **2**, the foregoing first leg assembly includes the first robotic thigh **111** and the first robotic calf **112**. The foregoing second leg assembly includes the second robotic thigh **211** and the second robotic calf **212**.

[0064] In some embodiments, the mobile robot further includes a first driving motor corresponding to the first wheel **113** and a second driving motor corresponding to the second wheel **213**. The first driving motor is configured to drive the first wheel **113** to rotate. The second driving motor is configured to drive the second wheel **213** to rotate.

[0065] In some embodiments, a plurality of first driving motor respectively corresponding to the plurality of first swinging legs **11** can control the plurality of first wheel **113** to rotate associatively or independently. In some embodiments, a plurality of second driving motor respectively corresponding to the plurality of second swinging legs **21** can control the plurality of second wheel **213** to rotate associatively or independently.

[0066] In some embodiments, the mobile robot further includes a waist structure **30** and a torso structure **40**. The waist structure **30** is configured to connect a leg structure and the torso structure **40**. The leg structure includes the first swinging leg set **10** and the second swinging leg set **20**.

[0067] In some embodiments, the waist structure **30** includes a pitch rotation axis **3**. The pitch rotation axis **3** is parallel to a rotation axis of the first swinging leg set **10**, and/or the pitch rotation axis **3** is parallel to a rotation axis of the second swinging leg set **20**. With reference to FIG. **1** and FIG. **2**, the pitch rotation axis **3** is parallel to the first hip rotation axis **1** (to the second hip rotation axis **2**). The pitch rotation axis **3** is rotatably connected to the torso structure **40**. The pitch rotation axis **3** is configured to support the torso structure **40** to perform a pitching operation.

[0068] In some embodiments, the waist structure **30** includes a sideway swinging rotation axis **4**. The sideway swinging rotation axis **4** is perpendicular to the rotation axis of the first swinging leg set **10**, and/or the sideway swinging rotation axis **4** is perpendicular to the rotation axis of the second swinging leg set **20**. With reference to FIG. **1** and FIG. **2**, the sideway swinging rotation axis **4** is perpendicular to the first hip rotation axis **1** (to the second hip rotation axis **2**). The sideway swinging rotation axis **4** is rotatably connected to the torso structure **40**, and. The sideway swinging rotation axis **4** is configured to support the torso structure **40** to perform a side swing operation.

[0069] In some embodiments, the waist structure **30** includes the pitch rotation axis **3** and the sideway swinging rotation axis **4**. The pitch rotation axis **3** is parallel to a rotation axis of the first swinging leg set **10**, and/or the pitch rotation axis **3** is parallel to a rotation axis of the second swinging leg set **20**. With reference to FIG. **1** and FIG. **2**, the pitch rotation axis **3** is parallel to the first hip rotation axis **1** (to the second hip rotation axis **2**). The pitch rotation axis **3** is configured to support the torso structure **40** to perform the pitching operation. The sideway swinging rotation

axis **4** is perpendicular to the pitch rotation axis **3**. A first end **41** of the sideway swinging rotation axis **4** is connected to a center of the pitch rotation axis **3**, a second end **42** of the sideway swinging rotation axis **4** is connected to the torso structure **40**, and the sideway swinging rotation axis **4** is configured to support the torso structure **40** to perform the side swing operation.
[0070] In some embodiments, the mobile robot further has at least one motion arm **50**. With reference to FIG. **1** and FIG. **2**, FIG. **1** and FIG. **2** show that the mobile robot has two motion arms **50**. The two motion arms **50** are symmetrically distributed along a central axis of the robot.
[0071] In some embodiments, the motion arm **50** is connected to a shoulder rotation axis **5** of the mobile robot. The shoulder rotation axis **5** is configured to support the motion arm **50** to have a multi-directional rotation degree of freedom. In some embodiments, the shoulder rotation axis **5** supports the motion arm **50** to rotate within a rotation angle range allowed by a structure of the robot. In some embodiments, the mobile robot further includes a shoulder driving motor corresponding to the shoulder rotation axis **5**. The shoulder driving motor is configured to drive the motion arm **50** to rotate. In some embodiments, a plurality of shoulder driving motor respectively corresponding to a plurality of shoulder rotation axes **5** can control a plurality of motion arms **50** to rotate associatively or independently.
[0072] In some embodiments, the motion arm **50** includes a robotic upper arm **51** and a robotic forearm **52**. The robotic upper arm **51** and the robotic forearm **52** are connected through an elbow joint rotation axis **6**. The elbow joint rotation axis **6** is configured to support the robotic forearm **52** to have a multi-directional rotation degree of freedom. In some embodiments, the elbow joint rotation axis **6** supports the robotic forearm **52** to rotate within a rotation angle range allowed by the structure of the robot. In some embodiments, the mobile robot further includes an elbow joint driving motor corresponding to the elbow joint rotation axis **6**. The wrist joint driving motor is configured to drive the robotic forearm **52** to rotate. In some embodiments, a plurality of wrist joint driving motors respectively corresponding to a plurality of elbow joint rotation axes **6** can control a plurality of robotic forearms **52** to rotate associatively or independently.
[0073] In some embodiments, a claw is connected to an end of the robotic forearm **52**. In some embodiments, the mobile robot further has a head **60**. The head **60** is located above the torso structure **40**.
[0074] FIG. **3** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **3** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0075] Operation **320**: Control the mobile robot to be in an initial standing state.
[0076] In some embodiments, the mobile robot includes a first swinging leg set and a second swinging leg set. The first swinging leg set includes a plurality of first swinging legs and/or the second swinging leg set includes a plurality of second swinging legs. The first swinging leg set and the second swinging leg set are arranged side by side, and a rotation axis of the first swinging leg set and a rotation axis of the second swinging leg set are located on the same vertical plane. At least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs. In some embodiments, the first swinging leg set includes the plurality of first swinging legs, and the second swinging leg set includes one second swinging leg. In some embodiments, the first swinging leg set includes one first swinging leg, and the second swinging leg set includes the plurality of second swinging legs. In some embodiments, the first swinging leg set includes the plurality of first swinging legs, and the second swinging leg set includes the plurality of second swinging legs. In some embodiments, the plurality of first swinging legs and the plurality of second swinging legs are alternately distributed one by one. In some embodiments, the plurality of first swinging legs are distributed on two sides of the plurality of second swinging legs.
[0077] The initial standing state is an initial reference state in which the mobile robot executes a folding action method shown in FIG. **3**. Before performing a subsequent folding action, the mobile

robot may be in the initial standing state.
[0078] In some embodiments, the initial standing state includes a state in which the first swinging leg set and the second swinging leg set of the mobile robot are closed together and jointly stand for support. Closed together in the initial standing state means that sides of the first swinging leg set are attached to (or face) sides of the second swinging leg set in the standing state, or closed together in the initial standing state means that an included angle does not exist between a vertical axis of the first swinging leg set and a vertical axis of the second swinging leg set in the standing state.
[0079] In some embodiments, the initial standing state includes a state in which the first swinging leg set and the second swinging leg set of the mobile robot are closed together and only one swinging leg set stands for support. In some embodiments, the leg set supporting for support is the first swinging leg set or the second swinging leg set. Closed together in the initial standing state means that sides of the first swinging leg set are attached to (or face) sides of the second swinging leg set in the standing state, or closed together in the initial standing state means that an included angle does not exist between a vertical axis of the first swinging leg set and a vertical axis of the second swinging leg set in the standing state.
[0080] In some embodiments, the initial standing state includes a state in which the first swinging leg set and the second swinging leg set of the mobile robot stand with legs crossed. In some embodiments, in the state of standing with legs crossed, the first swinging leg set is located before or behind the second swinging leg set.
[0081] With reference to FIG. **4**, part (A) of FIG. **4** shows a schematic diagram in which an initial standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support. FIG. **4** shows that the first swinging leg set **10** is located on an outer side of the second swinging leg set **20**. A plurality of first swinging legs **11** of the first swinging leg set **10** are symmetrically distributed along a central axis of the mobile robot, and a plurality of second swinging legs **21** of the second swinging leg set **20** are symmetrically distributed along a central axis of the mobile robot. A distance between any first swinging leg **11** and the central axis of the mobile robot is greater than a distance between any second swinging leg **21** and the central axis of the mobile robot. FIG. **4** includes two first swinging legs **11** and two second swinging legs **21**. [0082] Operation **340**: Control the mobile robot to transform from the initial standing state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.
[0083] The folded state refers to a state in which the mobile robot is in a non-operating mode. A space volume occupied by the robot in the non-operating mode is smaller than a space volume occupied by the robot in an operating mode. The folded state is a state obtained by the robot by performing the folding action in the initial standing state.
[0084] The folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together. Horizontally closed together means that extending directions of the first swinging leg set and the second swinging leg set are parallel to a horizontal reference plane and the sides of the first swinging leg set is attached to (or face) the sides of the second swinging leg set. Horizontally closed together means that extending directions of the first swinging leg set and the second swinging leg set are parallel to the horizontal reference plane, and an included angle does not exist between a horizontal axis of the first swinging leg set and a horizontal axis of the second swinging leg set.
[0085] In some embodiments, the mobile robot further includes a torso structure. The folded state further includes a state in which the torso structure is attached to the first swinging leg set and/or the second swinging leg set after being folded. A front side or a back side of the torso structure is attached to a top side of the first swinging leg set and/or the second swinging leg set. With reference to FIG. **4**, part (B) of FIG. **4** shows a mobile robot in a folded state. Part (B) of FIG. **4** shows a state in which the first swinging leg set **10** and the second swinging leg set **20** are

horizontally closed together, and a front side of a torso structure is attached to top sides of the first swinging leg set **10** and the second swinging leg set **20**.

[0086] Based on the above, in the operating state, the first swinging leg set and the second swinging leg set of the mobile robot are arranged side by side. In a non-operating state, the first swinging leg set and the second swinging leg set of the mobile robot are horizontally closed together, so that a volume of the mobile robot in the non-operating state can be reduced.

[0087] FIG. **5** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **5** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0088] Operation **520**: Control the mobile robot to be in an initial standing state.

[0089] The initial standing state is an initial reference state in which the mobile robot executes a folding action method shown in FIG. **5**. Before performing a subsequent folding action, the mobile robot may be in the initial standing state.

[0090] With reference to FIG. **6**, part (A) of FIG. **6** shows a schematic diagram in which an initial standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support. [0091] Operation **540**: Control the mobile robot to transform from the initial standing state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split.

[0092] The horizontally split state is a state in which an extending direction of the first swinging leg set is opposite to an extending direction of the second swinging leg set. In the horizontally split state, an included angle of 180 degrees exists between a direction from a leg root to a leg end of the first swinging leg set and a direction from the leg root to a leg end of the second swinging leg set. With reference to FIG. **6**, part (B) of FIG. **6** shows that the first swinging leg set **10** and the second swinging leg set **20** are in a horizontally split state.

[0093] In some embodiments, the mobile robot is controlled to perform a splitting action in the initial standing state, until the mobile robot is in the horizontally split state. The splitting action is an action of gradually increasing an included angle between the first swinging leg set and the second swinging leg set. In some embodiments, the splitting action is an action of simultaneously controlling the first swinging leg set and the second swinging leg set to rotate in opposite directions. In some embodiments, the first swinging leg set is controlled to rotate clockwise along the hip rotation axis, and the second swinging leg set is controlled to rotate counterclockwise along the hip rotation axis. In some embodiments, the first swinging leg set is controlled to rotate counterclockwise along the hip rotation axis, and the second swinging leg set is controlled to rotate clockwise along the hip rotation axis.

[0094] In some embodiments, the splitting action is an action of controlling rotation of only one swinging leg set. In some embodiments, the first swinging leg set or the second swinging leg set is controlled to rotate clockwise along the hip rotation axis. In some embodiments, the first swinging leg set or the second swinging leg set is controlled to rotate counterclockwise along the hip rotation axis.

[0095] Part (D) of FIG. **6** shows an intermediate state in which the mobile robot performs a splitting action to reach a horizontally split state. An included angle exists between a first swinging leg set **10** and a second swinging leg set **20**. A splitting action corresponding to part (D) of FIG. **6** is an action of simultaneously controlling the first swinging leg set **10** and the second swinging leg set **20** to rotate in opposite directions.

[0096] In some embodiments, a first swinging leg includes a first leg and a first wheel located on an end of the first leg, and a second swinging leg includes a second leg and a second wheel located on an end of the second leg. A first protrusion is arranged at a root of the first leg, and a second protrusion is arranged at a root of the second leg.

[0097] After operation **540**, the method further includes: determining a plurality of first wheels and a plurality of first protrusions respectively corresponding to the plurality of first swinging legs and a plurality of second wheels and a plurality of second protrusions respectively corresponding to the plurality of second swinging legs as a support portion of the mobile robot in the horizontally split state; and controlling the mobile robot to be in the horizontally split state. A position of the protrusion is represented by using a "triangle" in part (B) of FIG. **6**. [0098] Operation **560**: Control the mobile robot to transform from the horizontally split state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.

[0099] In some embodiments, the first swinging leg set or the second swinging leg set of the mobile robot is controlled to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together. The first swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the second swinging leg set as support legs in the horizontally split state, until the first swinging leg set and the second swinging leg set are horizontally closed together. When the leg rotation action is performed about the hip rotation axis, the first swinging leg set transforms in height.

[0100] The second swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the first swinging leg set as support legs in the horizontally split state, until the first swinging leg set and the second swinging leg set are horizontally closed together. When the leg rotation action is performed about the hip rotation axis, the second swinging leg set transforms in height.

[0101] In some embodiments, the mobile robot further includes the torso structure. The folded state further includes a state in which a front side or a back side of the torso structure is attached to the first swinging leg set and/or the second swinging leg set after being folded. The front side or the back side of the torso structure is oriented perpendicular to an extending direction of the first swinging leg set and/or the second swinging leg set. With reference to FIG. **6**, part (C) of FIG. **6** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together. A front side of the torso structure is attached to top sides of the first swinging leg set **10** and the second swinging leg set **20**.

[0102] Based on the above, a folding method is provided by controlling the mobile robot to be transformed from the initial standing state to the horizontally split state, and then controlling the mobile robot to be transformed from the horizontally split state to the horizontally closed state. Transformation of the robot from the initial standing state to the horizontally split state is implemented through the splitting action, which provides a self-folding method of the robot. In a process of implementing a complete folding operation, the robot may not be assisted by an external force, such as manpower, and the robot is always in a stable state in a folding process.

[0103] Based on some embodiments shown in FIG. **5**, if the mobile robot further includes the torso structure, the mobile robot further may perform an action of folding the torso structure. Operation **560** further includes operations shown in FIG. **7** in addition to "controlling the first swinging leg set or the second swinging leg set of the mobile robot to perform a leg rotation action about the hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together". Operations in FIG. **7** may be performed before the robot performs the leg rotation action, or may be performed after the robot performs the leg rotation action. The method shown in FIG. **7** includes the following operations. [0104] Operation **710**: Control the torso structure to perform a bending operation.

[0105] The torso structure is controlled to perform the bending operation through a pitch rotation axis.

[0106] In some embodiments, before the folding operation on the torso structure is performed, the torso structure may be controlled to be in an initial upright state. The initial upright state is a reference upright state of the torso structure. Before the folding operation is performed on the torso structure, the initial upright state may be restored to a reference upright state. In some embodiments, in the initial upright state, an orientation of a front side of the torso structure is parallel to an extending direction of the first swinging leg set in the horizontally split state, and/or an orientation of a front side of the torso structure is parallel to an extending direction of the second swinging leg set in the horizontally split state. In the initial upright state, a central axis of the torso structure is parallel to a gravity direction.

[0107] In some embodiments, before the folding operation on the torso structure is performed, the torso structure may not be controlled to be in a preset upright state, and the bending operation is performed by detecting a bending inclination angle of the torso structure. For example, before the folding operation is performed, the central axis of the torso structure of the mobile robot has an included angle with the gravity direction, and the bending operation is directly performed based on the included angle. [0108] Operation **720**: Control the front side or the back side of the torso structure to attach to a top side of the first swinging leg set; and/or control the front side or the back side of the torso structure to attach to a top side of the second swinging leg set.

[0109] The bending operation is performed until the front side or the back side of the torso structure attaches to the top side of the first swinging leg set, and it is determined that the torso structure completes the folding operation. An orientation of the front side or the back side of the torso structure in an attaching state is perpendicular to an extending direction of the first swinging leg set in the horizontally split state; and/or the bending operation is performed until the front side or the back side of the torso structure is attached to the top side of the second swinging leg set, and it is determined that the torso structure completes the folding operation. In the attaching state, an orientation of the front side or the back side of the torso structure in an attaching state is perpendicular to an extending direction of the second swinging leg set in the horizontally split state.

[0110] With reference to FIG. **8**, FIG. **8** shows a case in which a mobile robot first performs a folding operation on a torso structure before a leg rotation action is performed. Part (A) of FIG. **8**, part (B) of FIG. **8**, and part (C) of FIG. **8** respectively correspond to part (A) of FIG. **6**, part (B) of FIG. **6**, and part (C) of FIG. **6** that are introduced in FIG. **6**. Part (D) of FIG. **8** is a schematic diagram showing that a torso structure maintains being folded. A front side of a torso structure **40** is attached to a top side of a second swinging leg set **20**.

[0111] Based on some embodiments shown in FIG. **7**, the method in FIG. **7** may be performed before the mobile robot performs the leg rotation action, or may be performed after the mobile robot performs the leg rotation action. In a case in which the method in FIG. **7** is after the leg rotation action is performed (for example, a case in which the mobile robot performs a folding operation on a torso structure is after a folding operation is performed on a leg), before the folding operation on the leg is performed, the method further includes at least one of the following three operations. [0112] S1: Control the torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.

[0113] In some embodiments, the first swinging leg set is an outer swinging leg set, and the second swinging leg set is an inner swinging leg set. A distance between any swinging leg in the outer swinging leg set and a central axis of the mobile robot is greater than a distance between any swinging leg in the inner swinging leg set and the central axis of the mobile robot.

[0114] When the outer swinging leg set is to be rotated about a hip rotation axis by using the inner swinging leg set as support legs, before the leg rotation operation is performed, the torso structure is controlled to be in the sideway swinging state. A sideway swinging angle of the sideway swinging state is associated with a structure of the mobile robot. In some embodiments, the sideway swinging angle is a second included angle between an orientation of the front side of the torso structure and an extending direction of the inner swinging leg set in a horizontally split state.

In some embodiments, the sideway swinging angle is a first included angle between the orientation of the front side of the torso structure and an extending direction of the outer swinging leg set. With reference to FIG. **9**, FIG. **9** shows a first included angle θ.sub.1 between an orientation of a front side of a torso structure **40** and an extending direction of a first swinging leg set **10** in a sideway swinging state. FIG. **9** further shows a second included angle θ.sub.2 between an orientation of a front side of a torso structure **40** and an extending direction of a second swinging leg set **20** in a sideway swinging state.

[0115] In some embodiments, the sideway swinging angle is 30 degrees. When the sideway swinging angle is 30 degrees, interference does not occur between mechanical structures during the rotation of the outer swinging leg set. The outer swinging leg set can smoothly rotate about the hip rotation axis to a state in which the outer swinging leg set and the inner swinging leg set are horizontally closed together. In some embodiments, the sideway swinging angle is 90 degrees, which is a preset maximum avoidance angle, to ensure, to the greatest extent, that the outer swinging leg set does not interfere with the torso structure during the rotation. [0116] S2: Control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.

[0117] In some embodiments, before the leg rotation operation is performed, the torso structure is controlled to be in the bent state. A bending angle of the bent state is associated with a position of a center of gravity of the mobile robot. During execution of the leg rotation operation, a swinging leg set will be lifted. The torso structure is controlled to be in the bent state, so that the center of gravity of the robot always falls within a contact plane between the support leg set and a reference plane during the execution of the leg rotation operation. With reference to FIG. **10**, FIG. **10** shows a mobile robot in both a sideway swinging state and a bent state. FIG. **10** shows a third included angle θ.sub.3 between a central axis of a torso structure **40** and a gravity direction G in a bent state.

[0118] S3: Control a motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg set, the support leg set including a swinging leg set that comes into contact with a reference plane when a leg rotation action is performed.

[0119] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. Before the leg rotation operation is performed, the motion arm is controlled to be in the extended state in which the motion arm extends in the direction of the support leg set. A lifting height of the motion arm in the extended state is associated with the position of the center of gravity of the mobile robot. During the execution of the leg rotation operation, a swinging leg set will be lifted. The motion arm is controlled to stretch in the direction of the support leg set, so that the center of gravity of the robot always falls within the contact plane between the support leg set and the reference plane during the execution of the leg rotation operation.

[0120] With reference to FIG. **11**, FIG. **11** is a schematic diagram of a torso structure in a sideway swinging state and a bent state, with a motion arm being in a first extended state. FIG. **11** further shows a lifting height h of a motion arm **50**. A support leg set is a second swinging leg set **20**, and the motion arm **50** extends along a direction of the second swinging leg set **20**.

[0121] Operation S1, operation S2, and operation S3 above may be freely combined to a plurality of cases. In a case, before a leg rotation action is performed, only the torso structure is simultaneously controlled to be in the sideway swinging state and the bent state. In another case, before the leg rotation action is performed, only the torso structure is simultaneously controlled to be in the sideway swinging state and the motion arm is controlled to be in the first extended state. Various cases that can be obtained by combination are not enumerated herein. Somesome embodiments in which operation S1, operation S2, and operation S3 are simultaneously performed is described.

[0122] FIG. **12** is a flowchart of a method for controlling a mobile robot according to some

embodiments. The control method shown in FIG. **12** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0123] Operation **1201**: Control the mobile robot to be in an initial standing state.
[0124] The initial standing state is an initial reference state in which the mobile robot executes a folding action method shown in FIG. **12**. Before performing a subsequent folding action, the mobile robot may be in the initial standing state.
[0125] With reference to FIG. **13**, part (A) of FIG. **13** shows a schematic diagram in which an initial standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support. [0126] Operation **1202**: Control the mobile robot to transform from the initial standing state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split.
[0127] The horizontally split state is a state in which an extending direction of the first swinging leg set is opposite to an extending direction of the second swinging leg set. In the horizontally split state, an included angle of 180 degrees exists between a direction from a leg root to a leg end of the first swinging leg set and a direction from the leg root to a leg end of the second swinging leg set. With reference to FIG. **13**, part (B) of FIG. **13** shows that the first swinging leg set **10** and the second swinging leg set **20** are in a horizontally split state. [0128] Operation **1203**: Control a torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.
[0129] In some embodiments, the first swinging leg set is an outer swinging leg set, and the second swinging leg set is an inner swinging leg set. A distance between any swinging leg in the outer swinging leg set and a central axis of the mobile robot is greater than a distance between any swinging leg in the inner swinging leg set and the central axis of the mobile robot.
[0130] When the outer swinging leg set is to be rotated about a hip rotation axis by using the inner swinging leg set as support legs, before the leg rotation operation is performed, the torso structure is controlled to be in the sideway swinging state. A sideway swinging angle of the sideway swinging state is associated with a structure of the mobile robot. In some embodiments, the sideway swinging angle is a second included angle between an orientation of the front side of the torso structure and an extending direction of the inner swinging leg set in a horizontally split state. In some embodiments, the sideway swinging angle is a first included angle between the orientation of the front side of the torso structure and an extending direction of the outer swinging leg set.
[0131] In some embodiments, the sideway swinging angle is 30 degrees. When the sideway swinging angle is 30 degrees, interference does not occur between mechanical structures during the rotation of the outer swinging leg set. The outer swinging leg set can smoothly rotate about the hip rotation axis to a state in which the outer swinging leg set and the inner swinging leg set are horizontally closed together. In some embodiments, the sideway swinging angle is 90 degrees, which is a preset maximum avoidance angle, to ensure, to the greatest extent, that the outer swinging leg set does not interfere with the torso structure during the rotation.
[0132] Referring to FIG. **13**, part (C) of FIG. **13** shows that the torso structure **40** is in a sideway swinging state. [0133] Operation **1204**: Control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.
[0134] In some embodiments, before the leg rotation operation is performed, the torso structure is controlled to be in the bent state. A bending angle of the bent state is associated with a position of a center of gravity of the mobile robot. During execution of the leg rotation operation, a swinging leg

set will be lifted. The torso structure is controlled to be in the bent state, so that the center of gravity of the robot always falls within a contact plane between the support leg and a reference plane during the execution of the leg rotation operation. With reference to FIG. **13**, part (D) of FIG. **13** shows a mobile robot with a torso structure **40** being in a sideway swinging state and a bent state. [0135] Operation **1205**: Control a motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg, the support leg including a swinging leg set that comes into contact with a reference plane when a leg rotation action is performed.
[0136] In some embodiments, the mobile robot further has a motion arm located on a peripheral side of the torso structure. Before the leg rotation operation is performed, the motion arm is controlled to be in the extended state in which the motion arm extends in the direction of the support leg. A lifting height of the motion arm in the extended state is associated with the position of the center of gravity of the mobile robot. During the execution of the leg rotation operation, a swinging leg set will be lifted. The motion arm is controlled to stretch in the direction of the support leg, so that the center of gravity of the robot always falls within the contact plane between the support leg and the reference plane during the execution of the leg rotation operation. With reference to FIG. **13**, part (E) of FIG. **13** shows a mobile robot with a torso structure being in a sideway swinging state and a bent state and a motion arm being in a first extended state. [0137] Operation **1206**: Control the mobile robot to transform from the horizontally split state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.
[0138] In some embodiments, the first swinging leg set or the second swinging leg set of the mobile robot is controlled to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together. The first swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the second swinging leg set as support legs in the horizontally split state, until the first swinging leg set and the second swinging leg set are horizontally closed together. The second swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the first swinging leg set as support legs in the horizontally split state, until the first swinging leg set and the second swinging leg set are horizontally closed together. With reference to FIG. **13**, part (F) of FIG. **13** shows a mobile robot in a folded state.
[0139] Based on the above, before the leg rotation operation is performed, the mobile robot is controlled to be in the sideway swinging state, to avoid structural interference of the robot during the leg rotation. Before the leg rotation operation is performed, the mobile robot is controlled to be in the bent state and the motion arm is controlled to be in the first extended state, to resolve a problem that the robot is unstable as a result of an offset of the center of gravity during the leg rotation.
[0140] FIG. **14** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **14** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0141] Operation **1401**: Control the mobile robot to be in an initial standing state.
[0142] With reference to FIG. **15**, part (A) of FIG. **15** shows a schematic diagram in which an initial standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support. [0143] Operation **1402**: Control the mobile robot to perform the splitting action in the initial standing state, until the mobile robot is in a first split state, the first split state being a critical state in which the motion arm is supported on a reference plane.
[0144] During splitting, the mobile robot extends out of the motion arm in a gravity direction. The first split state is a critical state in which the motion arm is supported on the reference plane. Before

the critical state, the motion arm does not come into contact with the reference plane.
[0145] With reference to FIG. **15**, part (B) of FIG. **15** shows a critical state in which the motion arm **50** is supported on a reference plane. [0146] Operation **1403**: Control the mobile robot to further perform the splitting action until the mobile robot is in the horizontally split state.
[0147] In some embodiments, the mobile robot is controlled to further perform the splitting action, and control the motion arm of the mobile robot to continuously contract until the mobile robot is in the horizontally split state.
[0148] In some embodiments, in a process in which the mobile robot further performs the splitting action, the straight-strip-shaped motion arm is bent to a curved motion arm, to continuously contract the motion arm. By bending an elbow joint of the motion arm, the motion arm continuously contracts. Part (C) of FIG. **15** is a schematic diagram showing that an elbow joint of a motion arm **50** is in a bent state. Part (C) of FIG. **15** shows that the mobile robot is in a horizontally split state.
[0149] In some embodiments, in a process in which the mobile robot further performs the splitting action, the motion arm contracts the robotic upper arm and/or the robotic forearm, to continuously contract the motion arm. The motion arm includes a robotic upper arm and a robotic forearm. The robotic upper arm and the robotic forearm are connected through sleeving, and the robotic upper arm and/or the robotic forearm are controlled to contract in a sleeving direction, thereby continuously contracting the motion arm. [0150] Operation **1404**: Control the mobile robot to transform from the horizontally split state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.
[0151] With reference to FIG. **15**, part (D) of FIG. **15** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are in a horizontally closed state.
[0152] Based on the above, the mobile robot supports the reference plane through the motion arm during splitting. It can be avoided that the mobile robot is controlled to perform splitting only through a hip joint, thereby reducing force bearing on the hip joint and improving stability of the robot.
[0153] A detailed process of transformation from the initial standing state to the folded state by the mobile robot has been described above. The initial standing state is a preset starting state in which the mobile robot performs a folding operation. The mobile robot further may be controlled to transform from any standing state to the initial standing state.
[0154] FIG. **16** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **16** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0155] Operation **1601**: Control the mobile robot to perform an action of stretching or retracting a first swinging leg set upward in a first standing state, the first standing state including a state in which the first swinging leg set and a second swinging leg set are closed together and only the first swinging leg set stands for support.
[0156] The first standing state is any standing state of the first mobile robot. In the first standing state, only the first swinging leg set stands for support. The first swinging leg set is an outer swinging leg set. With reference to FIG. **17**, part (A**1**) of FIG. **17** shows a state in which a first swinging leg set **10** and a second swinging leg set **20** are closed together and only the first swinging leg set **10** stands for support.
[0157] By stretching or retracting the first swinging leg set upward, the state in which the first swinging leg set and the second swinging leg set jointly support for support may be achieved. The first swinging leg set includes a first robotic thigh and a first robotic calf. The first robotic thigh and the first robotic calf are connected through sleeving. The mobile robot is controlled to contract the first robotic thigh and/or the first robotic calf in a sleeving direction, so that the first swinging leg set can be contracted upward. [0158] Operation **1602**: Control the mobile robot to perform an action of stretching or retracting a second swinging leg set upward in a second standing state, the

second standing state including a state in which the first swinging leg set and the second swinging leg set are closed together and only the second swinging leg set stands for support.

[0159] The second standing state is any standing state of the second mobile robot. In the second standing state, only the second swinging leg set stands for support. The second swinging leg set is an inner swinging leg set.

[0160] By stretching or retracting the second swinging leg set upward, the state in which the first swinging leg set and the second swinging leg set jointly support for support may be achieved. The second swinging leg set includes a second robotic thigh and a second robotic calf. The second robotic thigh and the second robotic calf are connected through sleeving. The mobile robot is controlled to contract the second robotic thigh and/or the second robotic calf in a sleeving direction, so that the second swinging leg set can be stretched or retracted upward. [0161] Operation **1603**: Control the mobile robot to rotate the first swinging leg set and/or the second swinging leg set about the hip rotation axis in a third standing state, the third standing state including a state in which the first swinging leg set and the second swinging leg set stand with legs crossed in a front-to-back direction.

[0162] The third standing state is any standing state of the third mobile robot. In the third standing state, the first swinging leg set and the second swinging leg set stand with legs crossed in a front-to-back direction. Standing with legs crossed in a front-to-back direction means that a second included angle exists between an extending direction of the first swinging leg set and an extending direction of the second swinging leg set in a standing state (not in a parallel state). In some embodiments, with the robot standing with legs crossed in a front-to-back direction, the first swinging leg set is located before the second swinging leg set, or the first swinging leg set is located behind the second swinging leg set. With the robot standing with legs crossed in a front-to-back direction, stop points of the first swinging leg set are located before stop points of the second swinging leg set, or the stop points of the first swinging leg set are located behind the stop points of the second swinging leg set.

[0163] With reference to FIG. **17**, part (A**2**) of FIG. **17** shows that the first swinging leg set **10** and the second swinging leg set **20** are in a state of standing with legs crossed in a front-to-back direction. The first swinging leg set **10** is located before the second swinging leg set **20**, the first swinging leg set is an outer swinging leg set, and the second swinging leg set is an inner swinging leg set.

[0164] By rotating the first swinging leg set and/or the second swinging leg set about the hip rotation axis, the state in which the first swinging leg set and the second swinging leg set jointly support for support may be achieved. [0165] Operation **1604**: Control the mobile robot to be in an initial standing state, the initial standing state including a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support.

[0166] The initial standing state is an initial reference state in which the mobile robot executes a folding action method shown in FIG. **16**. Before performing a subsequent folding action, the mobile robot may be in the initial standing state.

[0167] With reference to FIG. **17**, part (B) of FIG. **17** shows a mobile robot in an initial standing state. The first swinging leg set **10** and the second swinging leg set **20** are closed together and jointly stand for support. [0168] Operation **1605**: Control the mobile robot to transform from the initial standing state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.

[0169] With reference to FIG. **17**, part (C) of FIG. **17** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together.

[0170] Based on the above, a method for transforming from a plurality of standing states to the initial standing state is provided, thereby providing a complete process of folding the mobile robot.

[0171] Related content of completing the folding operation performed by the robot is described above, and related content of performing the unfolding operation by the robot is described below.

[0172] FIG. **18** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **18** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0173] Operation **1820**: Control the mobile robot to be in a folded state, the folded state including at least a state in which a first swinging leg set and a second swinging leg set are horizontally closed together.

[0174] In some embodiments, the mobile robot includes a first swinging leg set and a second swinging leg set. The first swinging leg set includes a plurality of first swinging legs and/or the second swinging leg set includes a plurality of second swinging legs. The first swinging leg set and the second swinging leg set are arranged side by side, and a rotation axis of the first swinging leg set and a rotation axis of the second swinging leg set are located on the same vertical plane. At least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs. In some embodiments, the first swinging leg set includes a plurality of first swinging legs, and the second swinging leg set includes one second swinging leg. In some embodiments, the first swinging leg set includes one first swinging leg, and the second swinging leg set includes the plurality of second swinging legs. In some embodiments, the first swinging leg set includes the plurality of first swinging legs, and the second swinging leg set includes the plurality of second swinging legs. In some embodiments, the plurality of first swinging legs and the plurality of second swinging legs are alternately distributed one by one. In some embodiments, the plurality of first swinging legs are distributed on two sides of the plurality of second swinging legs.

[0175] The folded state refers to a state in which the mobile robot is in a non-operating mode. A space volume occupied by the robot in the non-operating mode is smaller than a space volume occupied by the robot in an operating mode. The folded state is a state obtained by the robot by performing a folding action in the target standing state.

[0176] The folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together. Horizontally closed together means that extending directions of the first swinging leg set and the second swinging leg set are parallel to a horizontal reference plane and the sides of the first swinging leg set is attached to (or face) the sides of the second swinging leg set. Horizontally closed together means that extending directions of the first swinging leg set and the second swinging leg set are parallel to the horizontal reference plane, and an included angle does not exist between a horizontal axis of the first swinging leg set and a horizontal axis of the second swinging leg set.

[0177] In some embodiments, the mobile robot further includes the torso structure. The folded state further includes a state in which a front side or a back side of the torso structure is attached to the first swinging leg set and/or the second swinging leg set after being folded. The front side or the back side of the torso structure is oriented perpendicular to an extending direction of the first swinging leg set and/or the second swinging leg set. With reference to FIG. **19**, part (A) of FIG. **19** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together. [0178] Operation **1840**: Control the mobile robot to transform from the folded state to the target standing state.

[0179] The target standing state is a reference unfolded state in which the mobile robot performs the unfolding method shown in FIG. **18**. The mobile robot may experience the target standing state to execute an unfolding action sequence.

[0180] In some embodiments, the target standing state includes a state in which the first swinging leg set and the second swinging leg set of the mobile robot are closed together and jointly stand for support. Closed together in the target standing state means that sides of the first swinging leg set are attached to (or face) sides of the second swinging leg set in the standing state, or closed together in the target standing state means that an included angle does not exist between a vertical axis of the first swinging leg set and a vertical axis of the second swinging leg set in the standing state.

[0181] In some embodiments, the target standing state includes a state in which a first swinging leg set and a second swinging leg set of the mobile robot are closed together and only one swinging leg set stands for support. In some embodiments, the leg set supporting for support is the first swinging leg set or the second swinging leg set. Closed together in the target standing state means that sides of the first swinging leg set are attached to (or face) sides of the second swinging leg set in the standing state, or closed together in the target standing state means that an included angle does not exist between a vertical axis of the first swinging leg set and a vertical axis of the second swinging leg set in the standing state.
[0182] In some embodiments, the target standing state includes a state in which the first swinging leg set and the second swinging leg set of the mobile robot stand with legs crossed. In some embodiments, in the state of standing with legs crossed, the first swinging leg set is located before or behind the second swinging leg set.
[0183] With reference to FIG. **19**, part (B) of FIG. **19** is a schematic diagram in which a target standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support.
[0184] Based on the above, in the operating state, the first swinging leg set and the second swinging leg set of the mobile robot are arranged side by side. In a non-operating state, the first swinging leg set and the second swinging leg set of the mobile robot are horizontally closed together, so that a volume of the mobile robot in the non-operating state can be reduced.
[0185] FIG. **20** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **20** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0186] Operation **2020**: Control the mobile robot to be in a folded state, the folded state including at least a state in which a first swinging leg set and a second swinging leg set are horizontally closed together.
[0187] In some embodiments, the mobile robot further includes the torso structure. The folded state further includes a state in which a front side or a back side of the torso structure is attached to the first swinging leg set and/or the second swinging leg set after being folded. The front side or the back side of the torso structure is oriented perpendicular to an extending direction of the first swinging leg set and/or the second swinging leg set. With reference to FIG. **21**, part (A) of FIG. **21** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together. [0188] Operation **2040**: Control the mobile robot to transform from the folded state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split.
[0189] The horizontally split state is a state in which an extending direction of the first swinging leg set is opposite to an extending direction of the second swinging leg set. In the horizontally split state, an included angle of 180 degrees exists between a direction from a leg root to a leg end of the first swinging leg set and a direction from the leg root to a leg end of the second swinging leg set.
[0190] With reference to FIG. **21**, part (B) of FIG. **21** shows a mobile robot in a horizontally split state. A first swinging leg set **10** and a second swinging leg set **20** are horizontally split.
[0191] In some embodiments, the first swinging leg set or the second swinging leg set of the mobile robot is controlled to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally split. The first swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the second swinging leg set as support legs in the horizontally closed state, until the first swinging leg set and the second swinging leg set are horizontally split. When the leg rotation action is performed about the hip rotation axis, the first swinging leg set transforms in height.
[0192] The second swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the first swinging leg set as support legs in the horizontally closed state, until the first swinging leg set and the second swinging leg set are horizontally split. When the leg

rotation action is performed about the hip rotation axis, the second swinging leg set transforms in height.

[0193] In some embodiments, a first swinging leg includes a first leg and a first wheel located on an end of the first leg, and a second swinging leg includes a second leg and a second wheel located on an end of the second leg. A first protrusion is arranged at a root of the first leg, and a second protrusion is arranged at a root of the second leg.

[0194] After operation **2040**, the method further includes: determining a plurality of first wheels and a plurality of first protrusions respectively corresponding to the plurality of first swinging legs and a plurality of second wheels and a plurality of second protrusions respectively corresponding to the plurality of second swinging legs as a support portion of the mobile robot in the horizontally split state; and controlling the mobile robot to be in the horizontally split state. A position of the protrusion is represented by using a “triangle” in part (B) of FIG. **21**. [0195] Operation **2060**: Control the mobile robot to transform from the horizontally split state to a target standing state.

[0196] The target standing state is a reference unfolded state in which the mobile robot performs the unfolding method shown in FIG. **20**. The mobile robot may experience the target standing state to execute an unfolding action sequence.

[0197] With reference to FIG. **21**, part (C) of FIG. **21** is a schematic diagram in which a target standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support.

[0198] In some embodiments, the mobile robot is controlled to perform a stand-up action in the horizontally split state, until the mobile robot is in the target standing state. The stand-up action is an action of gradually decreasing an included angle between the first swinging leg set and the second swinging leg set.

[0199] In some embodiments, the stand-up action is an action of simultaneously controlling the first swinging leg set and the second swinging leg set to rotate in opposite directions. In some embodiments, the first swinging leg set is controlled to rotate clockwise along the hip rotation axis, and the second swinging leg set is controlled to rotate counterclockwise along the hip rotation axis. In some embodiments, the first swinging leg set is controlled to rotate counterclockwise along the hip rotation axis, and the second swinging leg set is controlled to rotate clockwise along the hip rotation axis.

[0200] In some embodiments, the stand-up action is an action of controlling rotation of only one swinging leg set. In some embodiments, the first swinging leg set or the second swinging leg set is controlled to rotate clockwise along the hip rotation axis. In some embodiments, the first swinging leg set or the second swinging leg set is controlled to rotate counterclockwise along the hip rotation axis.

[0201] Part (D) of FIG. **21** shows an intermediate state when the mobile robot performs the stand-up action to reach the target standing state. An included angle exists between a first swinging leg set **10** and a second swinging leg set **20**. A stand-up action corresponding to part (D) of FIG. **21** is an action of simultaneously controlling the first swinging leg set **10** and the second swinging leg set **20** to rotate in opposite directions.

[0202] Based on the above, a folding method is further provided by controlling the mobile robot to transform from the horizontally closed state to the horizontally split state, and then controlling the mobile robot to transform from the horizontally split state to the target standing state. The robot is transformed from the horizontally split state to the target standing state through the stand-up action, which provides a self-unfolding method of the robot. In a process of implementing a complete unfolding operation, the robot may not be assisted by an external force, such as manpower, and the robot is always in a stable state in an unfolding process.

[0203] Based on some embodiments shown in FIG. **20**, if the mobile robot further includes the torso structure, the mobile robot further may perform an action of unfolding the torso structure. Operation **2060** further includes operations shown in FIG. **22** in addition to “controlling the first

swinging leg set or the second swinging leg set of the mobile robot to perform a leg rotation action about the hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together". Operations in FIG. **22** may be performed before the robot performs the leg rotation action, or may be performed after the robot performs the leg rotation action. The method shown in FIG. **22** includes the following operations. [0204] Operation **2210**: Control the front side or the back side of the torso structure to attach to a top side of the first swinging leg set; and/or control the front side or the back side of the torso structure to attach to a top side of the second swinging leg set.
[0205] With reference to FIG. **23**, FIG. **23** shows a case in which a leg rotation action of a mobile robot is first performed, and then an unfolding operation of a torso structure is performed. Part (A) of FIG. **23**, part (B) of FIG. **23**, and part (C) of FIG. **23** respectively correspond to part (A) of FIG. **21**, part (B) of FIG. **21**, and part (C) of FIG. **21** that are introduced in FIG. **21**. Part (D) of FIG. **23** is a schematic diagram showing that a torso structure maintains being folded. A front side of the torso structure **40** is attached to the top side of the second swinging leg set **20**. [0206] Operation **2220**: Control the torso structure to perform a sit-up operation.
[0207] The torso structure is controlled to perform the sit-up operation through the pitch rotation axis.
[0208] In some embodiments, the sit-up operation is performed until the torso structure is in a target upright state. The target upright state is a reference upright state of the torso structure, and after the unfolding operation is performed on the torso structure, the torso structure may be restored to the reference upright state. In some embodiments, in the target upright state, an orientation of a front side of the torso structure is parallel to an extending direction of the first swinging leg set in the horizontally split state, and/or an orientation of a front side of the torso structure is parallel to an extending direction of the second swinging leg set in the horizontally split state. In the initial upright state, a central axis of the torso structure is parallel to a gravity direction.
[0209] In some embodiments, the sit-up operation is performed without controlling the torso structure to reach the preset upright state. The sit-up operation is performed, so that the central axis of the torso structure of the mobile robot reaches any included angle with the gravity direction.
[0210] Based on some embodiments shown in FIG. **22**, the unfolding method in FIG. **22** may be performed after the mobile robot performs the leg rotation action, or may be performed before the mobile robot performs the leg rotation action. In a case in which the method in FIG. **22** is before the leg rotation action is performed (for example, a case in which the mobile robot performs an unfolding operation on the torso structure is before the folding operation is performed on a leg), after the unfolding operation on the torso structure is performed and before the leg unfolding operation is performed, the method further includes at least one of the following three operations.
[0211] S4: Control the torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.
[0212] In some embodiments, the first swinging leg set is an outer swinging leg set, and the second swinging leg set is an inner swinging leg set. A distance between any swinging leg in the outer swinging leg set and a central axis of the mobile robot is greater than a distance between any swinging leg in the inner swinging leg set and the central axis of the mobile robot.
[0213] When the outer swinging leg set is to be rotated about a hip rotation axis by using the inner swinging leg set as support legs, before the leg rotation operation is performed, the torso structure is controlled to be in the sideway swinging state. A sideway swinging angle of the sideway swinging state is associated with a structure of the mobile robot. In some embodiments, the sideway swinging angle is a second included angle between an orientation of the front side of the torso structure and an extending direction of the inner swinging leg set in a horizontally split state.

In some embodiments, the sideway swinging angle is a first included angle between the orientation of the front side of the torso structure and an extending direction of the outer swinging leg set. With reference to FIG. **9**, FIG. **9** shows a first included angle θ.sub.1 and a second included angle θ.sub.2 in a sideway swinging state.

[0214] In some embodiments, the sideway swinging angle is 30 degrees. When the sideway swinging angle is 30 degrees, interference does not occur between mechanical structures during the rotation of the outer swinging leg set. The outer swinging leg set can smoothly rotate about the hip rotation axis to a state in which the outer swinging leg set and the inner swinging leg set are horizontally closed together. In some embodiments, the sideway swinging angle is 90 degrees, which is a preset maximum avoidance angle, to ensure, to the greatest extent, that the outer swinging leg set does not interfere with the torso structure during the rotation. [0215] S5: Control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.

[0216] In some embodiments, before the leg rotation operation is performed, the torso structure is controlled to be in the bent state. A bending angle of the bent state is associated with a position of a center of gravity of the mobile robot. During execution of the leg rotation operation, a swinging leg set will be put down. The torso structure is controlled to be in the bent state, so that the center of gravity of the robot always falls within a contact plane between the support leg set and a reference plane during the execution of the leg rotation operation.

[0217] With reference to FIG. **10**, FIG. **10** shows a mobile robot in both a sideway swinging state and a bent state. FIG. **10** shows a third included angle θ.sub.3 between a central axis of a torso structure in a bent state and a gravity direction. [0218] S6: Control a motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg, the support leg including a swinging leg set that comes into contact with a reference plane when a leg rotation action is performed.

[0219] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. Before the leg rotation operation is performed, the motion arm is controlled to be in the extended state in which the motion arm extends in the direction of the support leg. A lifting height of the motion arm in the extended state is associated with the position of the center of gravity of the mobile robot. During the execution of the leg rotation operation, a swinging leg set will be put down. The motion arm is controlled to stretch in the direction of the support leg, so that the center of gravity of the robot always falls within the contact plane between the support leg and the reference plane during the execution of the leg rotation operation.

[0220] With reference to FIG. **11**, FIG. **11** is a schematic diagram of a torso structure in a sideway swinging state and a bent state, with a motion arm being in a first extended state. FIG. **11** further shows a lifting height h of the motion arm.

[0221] Operation S4, operation S5, and operation S6 above may be freely combined to a plurality of cases. In a case, before a leg rotation action is performed, only the torso structure is simultaneously controlled to be in the sideway swinging state and the bent state. In another case, before the leg rotation action is performed, only the torso structure is simultaneously controlled to be in the sideway swinging state and the motion arm is controlled to be in the first extended state. Various cases that can be obtained by combination are not enumerated herein. Some embodiments in which operation S4, operation S5, and operation S6 are simultaneously performed are described.

[0222] FIG. **24** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **24** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0223] Operation **2401**: Control the mobile robot to be in a folded state, the folded state including at least a state in which a first swinging leg set and a second swinging leg set are horizontally closed together. [0224] Operation **2402**: Control a torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.

[0225] In some embodiments, the first swinging leg set is an outer swinging leg set, and the second swinging leg set is an inner swinging leg set. A distance between any swinging leg in the outer swinging leg set and a central axis of the mobile robot is greater than a distance between any swinging leg in the inner swinging leg set and the central axis of the mobile robot.

[0226] When the outer swinging leg set is to be rotated about a hip rotation axis by using the inner swinging leg set as support legs, before the leg rotation operation is performed, the torso structure is controlled to be in the sideway swinging state. A sideway swinging angle of the sideway swinging state is associated with a structure of the mobile robot. In some embodiments, the sideway swinging angle is a second included angle between an orientation of the front side of the torso structure and an extending direction of the inner swinging leg set in a horizontally split state. In some embodiments, the sideway swinging angle is a first included angle between the orientation of the front side of the torso structure and an extending direction of the outer swinging leg set.

[0227] In some embodiments, the sideway swinging angle is 30 degrees. When the sideway swinging angle is 30 degrees, interference does not occur between mechanical structures during the rotation of the outer swinging leg set. The outer swinging leg set can smoothly rotate about the hip rotation axis to a state in which the outer swinging leg set and the inner swinging leg set are horizontally closed together. In some embodiments, the sideway swinging angle is 90 degrees, which is a preset maximum avoidance angle, to ensure, to the greatest extent, that the outer swinging leg set does not interfere with the torso structure during the rotation. [0228] Operation **2403**: Control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.

[0229] In some embodiments, before the leg rotation operation is performed, the torso structure is controlled to be in the bent state. A bending angle of the bent state is associated with a position of a center of gravity of the mobile robot. During execution of the leg rotation operation, a swinging leg set will be put down. The torso structure is controlled to be in the bent state, so that the center of gravity of the robot always falls within a contact plane between the support leg set and a reference plane during the execution of the leg rotation operation. [0230] Operation **2404**: Control a motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg set, the support leg set including a swinging leg set that comes into contact with a reference plane when a leg rotation action is performed.

[0231] In some embodiments, the mobile robot further has a motion arm located on a peripheral side of the torso structure. Before the leg rotation operation is performed, the motion arm is controlled to be in the extended state in which the motion arm extends in the direction of the support leg set. A lifting height of the motion arm in the extended state is associated with the position of the center of gravity of the mobile robot. During the execution of the leg rotation operation, a swinging leg set will be put down. The motion arm is controlled to stretch in the direction of the support leg set, so that the center of gravity of the robot always falls within the contact plane between the support leg set and the reference plane during the execution of the leg rotation operation. [0232] Operation **2405**: Control the mobile robot to transform from the folded state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split.

[0233] The horizontally split state is a state in which an extending direction of the first swinging leg set is opposite to an extending direction of the second swinging leg set. In the horizontally split state, an included angle of 180 degrees exists between a direction from a leg root to a leg end of the first swinging leg set and a direction from the leg root to a leg end of the second swinging leg set.

[0234] The first swinging leg set or the second swinging leg set of the mobile robot is controlled to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally split. The first swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the second swinging leg set as support legs in the folded state, until the first swinging leg set and the second swinging leg set are horizontally split. The second swinging leg set is controlled to perform the leg rotation action about the hip rotation axis by using the first swinging leg set as support legs in the horizontally closed state, until the first swinging leg set and the second swinging leg set are horizontally split. [0235] Operation **2406**: Control the mobile robot to transform from the horizontally split state to a target standing state.
[0236] The target standing state is a reference unfolded state in which the mobile robot performs the unfolding method shown in FIG. **24**. The mobile robot may experience the target standing state to execute an unfolding action sequence.
[0237] Based on the above, before the leg rotation operation is performed, the mobile robot is controlled to be in the sideway swinging state, to avoid structural interference of the robot during the leg rotation. Before the leg rotation operation is performed, the mobile robot is controlled to be in the bent state and/or the motion arm is controlled to be in the first extended state, to resolve a problem that the robot is unstable as a result of an offset of the center of gravity during the leg rotation.
[0238] FIG. **25** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **25** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0239] Operation **2501**: Control the mobile robot to be in a folded state, the folded state including at least a state in which a first swinging leg set and a second swinging leg set are horizontally closed together.
[0240] With reference to FIG. **26**, part (A) of FIG. **26** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together. [0241] Operation **2502**: Control the mobile robot to transform from the folded state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split.
[0242] The horizontally split state is a state in which an extending direction of the first swinging leg set is opposite to an extending direction of the second swinging leg set. In the horizontally split state, an included angle of 180 degrees exists between a direction from a leg root to a leg end of the first swinging leg set and a direction from the leg root to a leg end of the second swinging leg set.
[0243] With reference to FIG. **26**, part (B) of FIG. **26** shows a mobile robot in a horizontally split state. A first swinging leg set **10** and a second swinging leg set **20** are horizontally split. [0244] Operation **2503**: Control the mobile robot to perform the stand-up action in the horizontally split state until the mobile robot is in a first split state, the first split state being a critical state in which the motion arm is supported on the reference plane.
[0245] During standing up, the mobile robot extends out of the motion arm in a gravity direction. The first split state is a critical state in which the motion arm is supported on the reference plane. After the critical state, the motion arm does not come into contact with the reference plane.
[0246] In some embodiments, in a process in which the mobile robot performs the stand-up action, a curve-shaped motion arm is straightened to a straight-strip-shaped motion arm, so that the motion arm is continuously extended. The motion arm is constantly extended by straightening an elbow joint of the motion arm. Part (B) of FIG. **26** is a schematic diagram showing that an elbow joint of a motion arm **50** is in a bent state. Part (B) of FIG. **26** shows that the mobile robot is in a horizontally split state.
[0247] In some embodiments, in a process in which the mobile robot performs the stand-up action, the motion arm stretches a robotic upper arm and/or a robotic forearm, to continuously stretch the

motion arm. The motion arm includes a robotic upper arm and a robotic forearm. The robotic upper arm and the robotic forearm are connected through sleeving, and the robotic upper arm and/or the robotic forearm are/is controlled to stretch in a sleeving direction, to continuously stretch the motion arm.
[0248] With reference to FIG. **26**, part (C) of FIG. **26** shows a critical state in which the motion arm **50** is supported on a reference plane. Part (C) of FIG. **26** shows that the motion arm is in a straightened state. [0249] Operation **2504**: Control the mobile robot to further perform the stand-up action until the mobile robot is in the target standing state.
[0250] In some embodiments, the mobile robot is controlled to further perform the stand-up action, and control the motion arm of the mobile robot to continuously stretch until the mobile robot is in the target standing state.
[0251] The target standing state is a reference unfolded state in which the mobile robot performs the unfolding method shown in FIG. **25**. The mobile robot may experience the target standing state to execute an unfolding action sequence.
[0252] With reference to FIG. **26**, part (D) of FIG. **26** is a schematic diagram in which a target standing state is a state in which a first swinging leg set **10** and a second swinging leg set **20** of a mobile robot are closed together and jointly stand for support.
[0253] Based on the above, the mobile robot supports the reference plane through the motion arm during standing. It can be avoided that the mobile robot is controlled to perform standing only through a hip joint, thereby reducing force bearing on the hip joint and improving stability of the robot.
[0254] A detailed process in which the mobile robot transforms from the folded state to the target standing state has been described above. The target standing state is a preset end state in which the mobile robot performs an unfolding operation. The mobile robot further may be controlled to transform from the target standing state to any standing state.
[0255] FIG. **27** is a flowchart of a method for controlling a mobile robot according to some embodiments. The control method shown in FIG. **27** is performed by a controller of the mobile robot. The controller of the mobile robot may be located in a body of the robot or outside the robot. The method includes the following operations. [0256] Operation **2701**: Control the mobile robot to be in a folded state, the folded state including at least a state in which a first swinging leg set and a second swinging leg set are horizontally closed together.
[0257] With reference to FIG. **28**, part (A) of FIG. **28** shows a mobile robot in a folded state. The first swinging leg set **10** and the second swinging leg set **20** are horizontally closed together. [0258] Operation **2702**: Control the mobile robot to transform from the folded state to the target standing state, the target standing state including a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support.
[0259] The target standing state is a reference unfolded state in which the mobile robot performs the unfolding method shown in FIG. **27**. The mobile robot may experience the target standing state to execute an unfolding action sequence.
[0260] With reference to FIG. **28**, part (B) of FIG. **28** shows a mobile robot in a target standing state. The first swinging leg set **10** and the second swinging leg set **20** are in a state of closed together and jointly standing for support. [0261] Operation **2703**: Control the mobile robot to stretch the first swinging leg set downward in the target standing state.
[0262] The first swinging leg set is an outer swinging leg set. With reference to FIG. **28**, part (C**1**) of FIG. **28** shows a state in which a first swinging leg set **10** and a second swinging leg set **20** are closed together and only the first swinging leg set **10** stands for support, which is obtained by extending the first swinging leg set **10** downward.
[0263] By extending the first swinging leg set downward, a state in which only the first swinging leg set stands for support may be achieved. The first swinging leg set includes a first robotic thigh and a first robotic calf. The first robotic thigh and the first robotic calf are connected through

sleeving. The mobile robot is controlled to stretch the first robotic thigh and/or the first robotic calf in a sleeving direction, so that the first swinging leg set can be extended downward. [0264] Operation **2704**: Control the mobile robot to stretch the second swinging leg set downward in the target standing state.

[0265] The second swinging leg set is an inner swinging leg set. By extending the second swinging leg set downward, a state in which only the second swinging leg set stands for support may be achieved. The second swinging leg set includes a second robotic thigh and a second robotic calf. The second robotic thigh and the second robotic calf are connected through sleeving. The mobile robot is controlled to stretch the second robotic thigh and/or the second robotic calf in a sleeving direction, so that the second swinging leg set can be extended downward. [0266] Operation **2705**: Control the mobile robot to rotate the first swinging leg set and/or the second swinging leg set about the hip rotation axis in the target standing state, until the first swinging leg set and the second swinging leg set are in a state of standing with legs crossed in a front-to-back direction.

[0267] Standing with legs crossed in a front-to-back direction means that a second included angle exists between an extending direction of the first swinging leg set and an extending direction of the second swinging leg set in a standing state (not in a parallel state). In some embodiments, with the robot standing with legs crossed in a front-to-back direction, the first swinging leg set is located before the second swinging leg set, or the first swinging leg set is located behind the second swinging leg set. With the robot standing with legs crossed in a front-to-back direction, stop points of the first swinging leg set are located before stop points of the second swinging leg set, or the stop points of the first swinging leg set are located behind the stop points of the second swinging leg set.

[0268] With reference to FIG. **28**, part (C**2**) of FIG. **28** shows a first swinging leg set **10** and a second swinging leg set **20** are in a state of standing with legs crossed in a front-to-back direction. The first swinging leg set **10** is located before the second swinging leg set **20**, the first swinging leg set **10** is an outer swinging leg set, and the second swinging leg set **20** is an inner swinging leg set.

[0269] By rotating the first swinging leg set and/or the second swinging leg set about the hip rotation axis, a state in which the first swinging leg set and the second swinging leg set are in a state of standing with legs crossing.

[0270] Based on the above, a method for transforming from the target standing state to a plurality of random standing states is provided, thereby providing a complete unfolding process of the mobile robot.

[0271] Apparatus embodiments are described below. For additional implementation details, reference may be made to the corresponding descriptions of the above method embodiments.

[0272] FIG. **29** is a structural block diagram of an apparatus for controlling a mobile robot according to some embodiments. The mobile robot includes a first swinging leg set and a second swinging leg set. At least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs. The first swinging leg set and the second swinging leg set are arranged side by side, and rotation axes of the first swinging leg set and the second swinging leg set are located on the same vertical plane. The apparatus includes: [0273] a control module **2901**, configured to control the mobile robot to be in an initial standing state, [0274] the control module **2901** being further configured to control the mobile robot to transform from the initial standing state to a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together.

[0275] In some embodiments, the control module **2901** is further configured to control the mobile robot to transform from the initial standing state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split; and control the mobile robot to transform from the horizontally split state to the folded state.

[0276] In some embodiments, the control module **2901** is further configured to control the mobile robot to perform a splitting action in the initial standing state, until the mobile robot is in the horizontally split state.

[0277] In some embodiments, the control module **2901** is further configured to control the first swinging leg set or the second swinging leg set of the mobile robot to perform a leg rotation action about the hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together.

[0278] In some embodiments, the robot further has a torso structure. The control module **2901** is further configured to control the torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.

[0279] In some embodiments, the robot further has a torso structure. The control module **2901** is further configured to control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.

[0280] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. The control module **2901** is further configured to control a motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg set, the support leg set including a swinging leg set that comes into contact with the reference plane when the leg rotation action is performed.

[0281] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. The control module **2901** is further configured to: control the mobile robot to perform the splitting action in the initial standing state, until the mobile robot is in a first split state, the first split state being a critical state in which the motion arm is supported on a reference plane; and control the mobile robot to further perform the splitting action until the mobile robot is in the horizontally split state.

[0282] In some embodiments, the control module **2901** is further configured to: control the mobile robot to further perform the splitting action, and control the motion arm of the mobile robot to continuously contract until the mobile robot is in the horizontally split state.

[0283] In some embodiments, a first swinging leg includes a first leg and a first wheel located on an end of the first leg, and a second swinging leg includes a second leg and a second wheel located on an end of the second leg. A first protrusion is arranged at a root of the first leg, and a second protrusion is arranged at a root of the second leg. The control module **2901** is further configured to: determine a plurality of first wheels and a plurality of first protrusions respectively corresponding to the plurality of first swinging legs and a plurality of second wheels and a plurality of second protrusions respectively corresponding to the plurality of second swinging legs as a support portion of the mobile robot in the horizontally split state; and control the mobile robot to be in the horizontally split state.

[0284] In some embodiments, the mobile robot further includes the torso structure. The control module **2901** is further configured to: control the torso structure to perform a bending operation; control a front side or a back side of the torso structure to attach to a top side of the first swinging leg set; and control the front side or the back side of the torso structure to attach to a top side of the second swinging leg set.

[0285] In some embodiments, the initial standing state includes a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support. The control module **2901** is further configured to control the mobile robot to perform an action of stretching or retracting a first swinging leg set upward in a first standing state, the first standing state including a state in which the first swinging leg set and a second swinging leg set are closed together and only the first swinging leg set stands for support.

[0286] In some embodiments, the initial standing state includes a state in which the first swinging

leg set and the second swinging leg set are closed together and jointly stand for support. The control module **2901** is further configured to control the mobile robot to perform an action of stretching or retracting a second swinging leg set upward in a second standing state, the second standing state including a state in which the first swinging leg set and the second swinging leg set are closed together and only the second swinging leg set stands for support.
[0287] In some embodiments, the initial standing state includes a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support. The control module **2901** is further configured to control the mobile robot to rotate the first swinging leg set and/or the second swinging leg set about the hip rotation axis in a third standing state, the third standing state including a state in which the first swinging leg set and the second swinging leg set stand with legs crossed in a front-to-back direction.
[0288] Based on the above, in the operating state, the first swinging leg set and the second swinging leg set of the mobile robot are arranged side by side. In a non-operating state, the first swinging leg set and the second swinging leg set of the mobile robot are horizontally closed together, so that a volume of the mobile robot in the non-operating state can be reduced.
[0289] FIG. **30** is a structural block diagram of an apparatus for controlling a mobile robot according to some embodiments. The mobile robot includes a first swinging leg set and a second swinging leg set. At least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs. The first swinging leg set and the second swinging leg set are arranged side by side, and rotation axes of the first swinging leg set and the second swinging leg set are located on the same vertical plane. The apparatus includes: [0290] a control module **3001**, configured to control the mobile robot to be in a folded state, the folded state including at least a state in which the first swinging leg set and the second swinging leg set are horizontally closed together, [0291] the control module **3001** being further configured to control the mobile robot to transform from the folded state into a target standing state.
[0292] In some embodiments, the control module **3001** is further configured to control the mobile robot to transform from the folded state to a horizontally split state, the horizontally split state including a state in which the first swinging leg set and the second swinging leg set are horizontally split; and control the mobile robot to transform from the horizontally split state to a target standing state.
[0293] In some embodiments, the control module **3001** is further configured to control the first swinging leg set or the second swinging leg set of the mobile robot to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally split.
[0294] In some embodiments, the mobile robot further has the torso structure. The control module **3001** is further configured to control the torso structure to be in a sideway swinging state, the sideway swinging state meaning that a first included angle exists between an orientation of a front side of the torso structure and an extending direction of the first swinging leg set, and/or the sideway swinging state meaning that a second included angle exists between the orientation of the front side of the torso structure and an extending direction of the second swinging leg set.
[0295] In some embodiments, the mobile robot further has the torso structure. The control module **3001** is further configured to control the torso structure to be in a bent state, the bent state meaning that a third included angle exists between a central axis of the torso structure and a gravity direction.
[0296] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. The control module **3001** is further configured to control the motion arm to be in a first extended state, the first extended state being a state in which the motion arm extends in a direction of a support leg set, the support leg set including a swinging leg set that comes into contact with a reference plane when a leg rotation action is performed.
[0297] In some embodiments, the control module **3001** is further configured to control the mobile

robot to perform a stand-up action in the horizontally split state until the mobile robot is in the target standing state.

[0298] In some embodiments, the mobile robot further has a torso structure and a motion arm located on a peripheral side of the torso structure. The control module **3001** is further configured to: control the motion arm to be supported on a reference plane in the horizontally split state, and control the mobile robot to perform the stand-up action until the mobile robot is in a first split state, the first split state being a critical state in which the motion arm is supported on the reference plane; and control the mobile robot to further perform the stand-up action until the mobile robot is in the target standing state.

[0299] In some embodiments, the control module **3001** is further configured to: control the motion arm to be supported on the reference plane in the horizontally split state, control the motion arm to continuously extend, and control the mobile robot to perform the stand-up action until the mobile robot is in the first split state.

[0300] In some embodiments, a first swinging leg includes a first leg and a first wheel located on an end of the first leg, and a second swinging leg includes a second leg and a second wheel located on an end of the second leg. A first protrusion is arranged at a root of the first leg, and a second protrusion is arranged at a root of the second leg. The control module **3001** is further configured to: determine a plurality of first wheels and a plurality of first protrusions respectively corresponding to the plurality of first swinging legs and a plurality of second wheels and a plurality of second protrusions respectively corresponding to the plurality of second swinging legs as a support portion of the mobile robot in the horizontally split state; and control the mobile robot to be in the horizontally split state.

[0301] In some embodiments, the mobile robot further includes a torso structure. The control module **3001** is further configured to: control a front side or a back side of the torso structure to attach to a top side of the first swinging leg set; and/or control the front side or the back side of the torso structure to attach to a top side of the second swinging leg set; and control the torso structure to perform a sit-up operation.

[0302] In some embodiments, the target standing state includes a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support. The control module **3001** is further configured to: control the mobile robot to stretch the first swinging leg set downward in the target standing state; or control the mobile robot to stretch the second swinging leg set downward in the target standing state.

[0303] In some embodiments, the target standing state includes a state in which the first swinging leg set and the second swinging leg set are closed together and jointly stand for support. The control module **3001** is further configured to control the mobile robot to rotate the first swinging leg set and/or the second swinging leg set about the hip rotation axis in the target standing state, until the first swinging leg set and the second swinging leg set are in a state of standing with legs crossed in a front-to-back direction.

[0304] Based on the above, in the operating state, the first swinging leg set and the second swinging leg set of the mobile robot are arranged side by side. In a non-operating state, the first swinging leg set and the second swinging leg set of the mobile robot are horizontally closed together, so that a volume of the mobile robot in the non-operating state can be reduced.

[0305] According to some embodiments, each module may exist respectively or be combined into one or more modules. Some modules may be further split into multiple smaller function subunits, thereby implementing the same operations without affecting the technical effects of some embodiments. The modules are divided based on logical functions. In actual applications, a function of one module may be realized by multiple modules, or functions of multiple modules may be realized by one module. In some embodiments, the apparatus may further include other modules. In actual applications, these functions may also be realized cooperatively by the other modules, and may be realized cooperatively by multiple modules.

[0306] A person skilled in the art would understand that these “modules” could be implemented by hardware logic, a processor or processors executing computer software code, or a combination of both. The “modules” may also be implemented in software stored in a memory of a computer or a non-transitory computer-readable medium, where the instructions of each module are executable by a processor to thereby cause the processor to perform the respective operations of the corresponding module.
[0307] FIG. **31** is a structural block diagram of a mobile robot according to some embodiments. The mobile robot includes a controller **3101** and a memory **3102**.
[0308] The controller **3101** may include one or more processing cores, such as a 4-core processor and an 8-core processor. The controller **3101** may be implemented in at least one hardware form of a digital signal processor (DSP), a field-programmable gate array (FPGA), and a programmable logic array (PLA). The controller **3101** may include a main processor and a coprocessor. The main processor is configured to process data in a wake-up state, which is also referred to as a central processing unit (CPU). The coprocessor is a low-power processor configured to process data in a standby state. In some embodiments, the processor **3101** may have a graphics processing unit (GPU) integrated therein. The GPU is configured to render and draw content that may be displayed on a display screen. In some embodiments, the controller **3101** may further include an artificial intelligence (AI) processor. The AI processor is configured to perform computing operations related to machine learning. The memory **3102** may include one or more computer-readable storage media. The computer-readable storage medium may be non-transient. The memory **3102** may further include a high-speed random access memory and a nonvolatile memory, for example, one or more disk storage devices or flash storage devices. In some embodiments, the non-transient computer-readable storage medium in the memory **3102** is configured to store at least one instruction. The at least one instruction is executed by the controller **3101** to implement the method for controlling a mobile robot provided in the method embodiments in some embodiments.
[0309] In some embodiments, the mobile robot may include at least one motor **3103** and at least one sensor **3104**. The at least one motor **3103** is configured to receive a control instruction transmitted by the controller **3101**, and drive the mobile robot to perform actions. The at least one motor **3103** drives each joint of the mobile robot to perform actions such as rotation/stretching or retracting/fixing. The at least one sensor **3104** is configured to obtain state information of the mobile robot. The state information includes an internal state of the mobile robot and/or an external state (environment information) of the mobile robot. The at least one sensor **3104** transmits the status information of the mobile robot to the controller **3101**, to control the mobile robot to perform related actions.
[0310] A person skilled in the art may understand that the structure shown in FIG. **31** constitutes no limitation on the mobile robot, and may include more or fewer assemblies than illustrated, or combine some assemblies, or adopt different assembly arrangements.
[0311] Some embodiments further provide a mobile robot. The mobile robot includes a memory and a processor. The memory has at least one piece of program code stored therein, the program code is loaded and executed by the processor, to implement the foregoing method for controlling a mobile robot.
[0312] Some embodiments further provide a computer-readable storage medium, having a computer program stored therein, the computer program being executed by a processor to implement the foregoing method for controlling a mobile robot.
[0313] Some embodiments further provide a chip, including a programmable logic circuit and/or program instructions, the chip, when run, being configured to implement the foregoing method for controlling a mobile robot.
[0314] Some embodiments further provide a computer program product or a computer program, the computer program product or the computer program including computer instructions stored in a computer-readable storage medium, a processor reading the computer instructions from the

computer-readable storage medium and executing the computer instructions, to implement the foregoing method for controlling a mobile robot.

[0315] The foregoing embodiments are used for describing, instead of limiting the technical solutions of the disclosure. A person of ordinary skill in the art shall understand that although the disclosure has been described in detail with reference to the foregoing embodiments, modifications can be made to the technical solutions described in the foregoing embodiments, or equivalent replacements can be made to some technical features in the technical solutions, provided that such modifications or replacements do not cause the essence of corresponding technical solutions to depart from the spirit and scope of the technical solutions of the embodiments of the disclosure and the appended claims.

Claims

1. A method for controlling a mobile robot that includes a first swinging leg set and a second swinging leg set, wherein at least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane, the method comprising: controlling the mobile robot to be in an initial standing state; and controlling the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among the first swinging leg set and the second swinging leg set are horizontally closed together.

2. The method according to claim 1, wherein the controlling the mobile robot to transform from the initial standing state to the folded state comprises: controlling the mobile robot to transform from the initial standing state to a horizontally split state in which the first swinging leg set and the second swinging leg set are horizontally split; and controlling the mobile robot to transform from the horizontally split state to the folded state.

3. The method according to claim 2, wherein the controlling the mobile robot to transform from the initial standing state to the horizontally split state comprises: controlling the mobile robot to perform a splitting action in the initial standing state, until the mobile robot is in the horizontally split state.

4. The method according to claim 3, wherein the mobile robot includes a torso structure and a motion arm located on a peripheral side of the torso structure, and wherein the controlling the mobile robot to perform the splitting action in the initial standing state comprises: controlling the mobile robot to perform the splitting action in the initial standing state, until the mobile robot is in a first split state that is a critical state in which the motion arm is supported on a reference plane; and controlling the mobile robot to further perform the splitting action until the mobile robot is in the horizontally split state.

5. The method according to claim 4, wherein the controlling the mobile robot to further perform the splitting action comprises: controlling the mobile robot to further perform the splitting action, and controlling the motion arm of the mobile robot to continuously contract until the mobile robot is in the horizontally split state.

6. The method according to claim 2, wherein the controlling the mobile robot to transform from the horizontally split state to the folded state comprises: controlling one from among the first swinging leg set and the second swinging leg set to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together.

7. The method according to claim 6, wherein the mobile robot further includes a torso structure, and wherein the method further comprises: controlling the torso structure to be in a sideway swinging state, wherein: a first included angle exists between an orientation of a front side of the torso structure and a first extending direction of the first swinging leg set, a second included angle exists between the orientation of the front side of the torso structure and a second extending direction of the second swinging leg set, or the sideway swinging state meaning that the first included angle exists between the orientation of the front side of the torso structure and the first extending direction, and the second included angle exists between the orientation of the front side of the torso structure and the second extending direction.

8. The method according to claim 6, wherein the mobile robot includes a torso structure, and wherein the method further comprises: controlling the torso structure to be in a bent state, wherein a third included angle exists between a central axis of the torso structure and a gravity direction.

9. The method according to claim 6, wherein the mobile robot includes a torso structure and a motion arm located on a peripheral side of the torso structure, and wherein the method further comprises: controlling the motion arm to be in a first extended state, wherein the motion arm extends in a direction of a support leg set that includes a swinging leg set that comes into contact with a reference plane when the leg rotation action is performed.

10. The method according to claim 6, wherein the mobile robot includes a torso structure, and wherein the method further comprises: controlling the torso structure to perform a bending operation; and performing one from among: controlling one from among a front side and a back side of the torso structure to attach to a first top side of the first swinging leg set; controlling one from among the front side and the back side to attach to a second top side of the second swinging leg set; and controlling one from among the front side and the back side to attach to both the first top side and the second top side.

11. An apparatus for controlling a mobile robot, comprising: at least one memory configured to store computer program code; and at least one processor configured to read the program code and operate as instructed by the program code, wherein the mobile robot comprises a first swinging leg set and a second swinging leg set, wherein at least one of the first swinging leg set and the second swinging leg set comprises a plurality of swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane, and wherein the program code comprises: first control code configured to cause at least one of the at least on processor to control the mobile robot to be in an initial standing state; and second control code configured to cause at least one of the at least one processor to control the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among the first swinging leg set and the second swinging leg set are horizontally closed together.

12. The apparatus according to claim 11, wherein the second control code is configured to cause at least one of the at least on processor to: control the mobile robot to transform from the initial standing state to a horizontally split state in which the first swinging leg set and the second swinging leg set are horizontally split; and control the mobile robot to transform from the horizontally split state to the folded state.

13. The apparatus according to claim 12, wherein the second control code is configured to cause at least one of the at least on processor to: control the mobile robot to perform a splitting action in the initial standing state, until the mobile robot is in the horizontally split state.

14. The apparatus according to claim 13, wherein the mobile robot comprises a torso structure and a motion arm located on a peripheral side of the torso structure, and wherein the second control code is configured to cause at least one of the at least on processor to: control the mobile robot to perform the splitting action in the initial standing state, until the mobile robot is in a first split state that is a critical state in which the motion arm is supported on a reference plane; and control the mobile robot to further perform the splitting action until the mobile robot is in the horizontally split state.

15. The apparatus according to claim 14, wherein the second control code is configured to cause at least one of the at least on processor to: control the mobile robot to further perform the splitting action, and control the motion arm of the mobile robot to continuously contract until the mobile robot is in the horizontally split state.

16. The apparatus according to claim 12, wherein the second control code is configured to cause at least one of the at least on processor to: control one from among the first swinging leg set and the second swinging leg set to perform a leg rotation action about a hip rotation axis until the first swinging leg set and the second swinging leg set are horizontally closed together.
17. The apparatus according to claim 16, wherein the mobile robot further comprises a torso structure, and wherein the program code further comprises: third control code configured to cause at least one of the at least one processor to control the torso structure to be in a sideway swinging state, wherein: a first included angle exists between an orientation of a front side of the torso structure and a first extending direction of the first swinging leg set, a second included angle exists between the orientation of the front side of the torso structure and a second extending direction of the second swinging leg set, or the sideway swinging state meaning that the first included angle exists between the orientation of the front side of the torso structure and the first extending direction, and the second included angle exists between the orientation of the front side of the torso structure and the second extending direction.
18. The apparatus according to claim 16, wherein the mobile robot comprises a torso structure, and wherein the program code further comprises: third control code configured to cause at least one of the at least one processor to control the torso structure to be in a bent state, wherein a third included angle exists between a central axis of the torso structure and a gravity direction.
19. The apparatus according to claim 16, wherein the mobile robot comprises a torso structure and a motion arm located on a peripheral side of the torso structure, and wherein the program code further comprises: third control code configured to cause at least one of the at least one processor to control the motion arm to be in a first extended state, wherein the motion arm extends in a direction of a support leg set that includes a swinging leg set that comes into contact with a reference plane when the leg rotation action is performed.
20. A non-transitory computer-readable storage medium, storing computer code which, when executed by at least one processor, causes the at least one processor to at least: control a mobile robot to be in an initial standing state; and control the mobile robot to transform from the initial standing state to a folded state, wherein at least one from among a first swinging leg set of the mobile robot and a second swinging leg set of the mobile robot are horizontally closed together, wherein at least one of the first swinging leg set and the second swinging leg set includes a plurality of swinging legs, wherein the first swinging leg set and the second swinging leg set are arranged side by side, and wherein a first rotation axis of the first swinging leg set and a second rotation axis of the second swinging leg set are located on a same vertical plane.